Supplement 1. Why did Medicare ACOs reset their benchmarks early? A pre-registered analysis of early renewal in the Medicare Shared Savings Program

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All tables are generated from the committed results files by `scripts/make_tables.py` in the manuscript repository. Item and section numbers refer to the registered analysis plan (`PREREGISTRATION.md`, version 0.6).[1]

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eTable 1. Validation of the rebuilt benchmark against benchmarks CMS published for regular renewers

Check	ACOs, No.	Spearman ρ	Median absolute error, %	Slope	Intercept, $
Rebuilt 2022 benchmark, all	101	0.94	1.6	0.944	664
ACOs at the first phase-in step	62	0.93	2.0	0.917	886
ACOs at the second phase-in step	39	0.96	0.9	1.017	−47
Own-spending part (step 1) vs CMS benchmark-year spending	101	0.95	1.4	0.940	676

Validation sample: regular renewers whose 2022 benchmarks CMS publishes (101 of 111 eligible; the rest lack either the rebuilt or the published benchmark). Early renewers were excluded from the sample, and the 2022 results file was filtered as it was loaded. Registered pass rule: Spearman correlation at least 0.80 and median absolute percentage error at most 5%; calibration only if the slope of actual on rebuilt fell outside 0.9 to 1.1. Result: passed, no calibration. The error SD of the step-1 check sets the measurement-error SD of the own-spending component, $\sigma_v = 4.38$ pp.

eTable 2. Power analysis

Test	Spread used	Minimum detectable effect	Note
H1, savings rate	SD 4.9 pp (PY2016-PY2018)	2.7 pp	simulated power 80.4%
H2b, regional pull	SD 4.6 pp (stayers)	2.5 pp	0.55 SD
H2a, own-spending component	SD 3.3 pp (stayers)	1.8 pp	0.55 SD
H4, spending change	SD 6.9 log points (pre-period change)	3.8 log points	+3.9% or −3.7%
H3, regional pull	true log-odds difference 0.5 per 2 SD	detected in 17%	of 500 simulated datasets
H3, own-spending component	true log-odds difference 0.5 per 2 SD	detected in 0%	of 500 simulated datasets
H3, higher BASIC level	true log-odds difference 0.5 per 2 SD	detected in 14%	of 500 simulated datasets

Minimum detectable effects at 80% power, two-sided $\alpha = .05$, for 29 early renewers vs 277 stayers, computed after registration and committed before any confirmatory estimate; 1 log point = 0.01 in log spending. For H3, power was simulated only at a log-odds difference of 0.5; a predictor counts as detected when its 95% credible interval excludes zero.

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eTable 3. Robustness of the group comparisons

Analysis	Difference	95% interval	*P*
H2b regional pull			
Confirmatory	+3.4	1.7 to 5.4	<.001
Regression, cohort only (5.3)	+2.1	0.5 to 3.8	
Regression + track, level and size (item 1)	+2.2	0.7 to 3.8	
Difference in medians (item 2)	+2.8	0.1 to 5.2	<.001
2020-01-01 cohort only (item 3)	+3.7	1.8 to 5.5	.001
Unadjusted county spending (item 9)	+6.9	2.4 to 12.4	.002
H2a own-spending component			
Confirmatory	−2.1	−3.8 to −0.5	.003
Regression, cohort only (5.3)	−1.5	−2.8 to −0.3	
Regression + track, level and size (item 1)	−1.3	−2.6 to 0.0	
Difference in medians (item 2)[a]	−1.4	−3.2 to 0.0	.09
2020-01-01 cohort only (item 3)[a]	−1.8	−4.2 to 0.5	.04
BY2-BY3 only in step 1 (item 6)	−1.9	−3.7 to −0.2	.01
Complete cases (item 8)	−2.1	−3.9 to −0.5	.005
H1 savings rate, PY2020			
Confirmatory	+1.9	0.7 to 3.3	.01
Regression, cohort only (5.3)	+1.5	0.2 to 2.9	
Regression + track, level and size (item 1)	+1.2	−0.1 to 2.6	
Difference in medians (item 2)	+1.4	−0.1 to 3.3	.06
2020-01-01 cohort only (item 3)	+1.8	0.5 to 3.3	.03
Total expected gain			
Confirmatory	−0.5	−2.0 to 0.8	.33
Complete cases (item 8)	−0.5	−2.0 to 0.8	.33
Regional component			
Confirmatory	+1.6	0.9 to 2.2	<.001
Complete cases (item 8)	+1.6	0.9 to 2.2	<.001

Early renewers minus stayers, in the units of Table 2. Item numbers refer to the registered plan's robustness list (section 5.5); '5.3' is the registered covariate-adjusted estimate with cohort only. Regression rows are robust Bayesian linear regressions (Student-t errors on the standardized variable, Normal(0, 1) priors) of the variable on early renewal and cohort (5.3) or cohort, track, level and size (item 1), with 95% credible intervals and no P value; the robust likelihood downweights a few large values. Median rows give the bootstrap posterior mean of the difference in medians; their P

value tests the observed difference. [a]The P value and the 95% interval disagree about zero; the plan requires reporting both and noting the disagreement.

eTable 4. Robustness of the decision model (H3)

Analysis	Coefficient	95% interval	*P*
Regional pull			
Confirmatory	1.34	0.44 to 2.30	
Firth logistic regression (item 5)	1.33	0.25 to 2.54	.004
2020-01-01 cohort only (item 3)	1.15	−0.02 to 2.39	
BY2-BY3 only in step 1 (item 6)	1.33	0.42 to 2.26	
No measurement-error model (item 7)	1.22	0.34 to 2.20	
Complete cases (item 8)	1.34	0.38 to 2.32	
Unadjusted county spending (item 9)	0.75	−0.02 to 1.56	
Own-spending component			
Confirmatory	−0.05	−2.01 to 1.92	
Firth logistic regression (item 5)	−0.65	−1.74 to 0.30	.18
2020-01-01 cohort only (item 3)	−0.03	−1.97 to 1.96	
BY2-BY3 only in step 1 (item 6)	−0.02	−1.98 to 1.97	
No measurement-error model (item 7)	−0.64	−1.51 to 0.23	
Complete cases (item 8)	−0.06	−2.04 to 1.95	
Unadjusted county spending (item 9)	−0.04	−1.99 to 1.93	
Higher BASIC level (C-E)			
Confirmatory	1.34	0.38 to 2.31	
Firth logistic regression (item 5)	1.55	0.53 to 2.70	.006
2020-01-01 cohort only (item 3)	1.57	0.42 to 2.76	
BY2-BY3 only in step 1 (item 6)	1.33	0.41 to 2.28	
No measurement-error model (item 7)	1.26	0.33 to 2.23	
Complete cases (item 8)	1.34	0.41 to 2.32	
Unadjusted county spending (item 9)	1.52	0.61 to 2.47	

Log-odds differences per 2 SD (BASIC level: C-E vs A-B). The Firth regression includes the core predictors and cohort, with profile-likelihood intervals and a direct-simulation P value; it would have decided H3 only if the Bayesian model had failed its convergence checks.

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eTable 5. Convergence and predictive checks of the decision model (H3)

Check	Result
Largest R-hat (Normal / Student-t prior)	1.001 / 1.001
Smallest bulk effective sample size	6,661 / 6,179
Divergent transitions	0 / 0
Fits needed (registered maximum: 2)	1 / 2
Prior predictive share renewing early, median (95% range)	17% (1% to 70%)
Observed early renewers (posterior predictive 95% range)	24 (14 to 37)
Top predicted decile: observed vs predicted share	57% vs 45%
Direct-simulation refits not converged (pull / own / BASIC level)	0.34% / 0.38% / 0.43%
Overall test: largest absolute core coefficient	1.35, $P = .004$

eTable 6. Conventional (frequentist) counterparts

Hypothesis	Method	Estimate	95% CI	*P*
H2b, regional pull	Welch t test	+3.2	1.1 to 5.3	.004
	OLS, HC2 errors, cohort	+3.3	1.3 to 5.4	.001
	OLS, HC2 errors, adjusted	+3.0	0.9 to 5.1	.005
H2a, own-spending component	Welch t test	−2.1	−3.9 to −0.2	.03
	OLS, HC2 errors, cohort	−2.1	−3.8 to −0.4	.02
	OLS, HC2 errors, adjusted	−1.6	−3.3 to 0.1	.06
H1, savings rate PY2020	Welch t test	+1.8	0.4 to 3.3	.01
	OLS, HC2 errors, cohort	+1.9	0.5 to 3.3	.009
	OLS, HC2 errors, adjusted	+1.4	−0.1 to 2.8	.06
H4, event study (TWFE)	PY2020 vs PY2021	−0.7	−4.8 to 3.4	.73
	PY2022 vs PY2021	−0.4	−2.8 to 2.0	.75
	PY2023 vs PY2021	−2.6	−7.4 to 2.2	.29
	PY2024 vs PY2021	−3.1	−7.9 to 1.7	.20

Reported only; not used for any decision. Differences are early renewers minus stayers (pp; H1 in pp of savings rate). Adjusted OLS adds track, level and size to cohort. Event-study coefficients are log points (0.01 in log spending) relative to PY2021, from two-way fixed effects with errors clustered by ACO. TWFE, two-way fixed effects.

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eTable 7. Agreement of registered and conventional verdicts

Claim	Registered method	Conventional method	Agree
H2b	Supported	Supported	Yes
H2a	Contradicted	Contradicted	Yes
H3 regional pull	Supported	Supported	Yes
H3 own-spending component	Undetermined	Undetermined	Yes
H3 higher BASIC level	Supported	Supported	Yes
H4	Undetermined	Undetermined	Yes

For H2a and H2b the registered conventional counterpart is OLS with HC2 errors; for H3, maximum-likelihood logistic regression; for H4, two-way fixed effects. 'Undetermined' for H4 is the registered label for 'no detectable difference'.

eTable 8. Attrition from the H4 spending panel, by group and year

Group and status	PY2022	PY2023	PY2024
Early renewers (29): included	29	28	25
Early renewers (29): left the program	0	1	1
Early renewers (29): renewed (no longer 'not yet renewed')	0	0	3
Stayers (277): included	276	222	162
Stayers (277): left the program	0	46	72
Stayers (277): renewed (no longer 'not yet renewed')	0	8	42
Stayers (277): no spending value	1	1	1

Required by the registered plan (section 4.5). PY2020 and PY2021 include all 29 early renewers and 277 stayers. Follow-up: a mean of 2.83 included post-period years per early renewer and 2.39 per comparison ACO (of 3).

eTable 9. Exploratory analyses

Analysis	Estimate	95% CrI
H2b, early renewers minus stayers (pp of benchmark)		
Regional pull (confirmatory)	+3.4	1.7 to 5.4
Pull weighted by phase-in weight w	+1.6	0.7 to 2.6
Capped regional component	+1.6	0.9 to 2.2
H3 coefficient on the pull measure (log-odds)		

Analysis	Estimate	95% CrI
Regional pull (confirmatory)	+1.34	0.44 to 2.30
Pull weighted by phase-in weight w	+1.20	0.40 to 2.04
Capped regional component	+1.49	0.57 to 2.45
Census region added	+1.43	0.46 to 2.41
Census region in H3 (log-odds vs Northeast)		
Midwest vs Northeast	+0.35	−0.85 to 1.50
South vs Northeast	−0.12	−1.12 to 0.89
West vs Northeast	+0.09	−1.21 to 1.38

Labeled exploratory in the registered plan (section 5.6): no predictions and no verdicts. 2 of the 334 ACOs have no census region; the region model's sample is 206 ACOs (24 early renewers), as in H3.

eTable 10. The 2 early renewers already in ENHANCED

Variable	ACO 1 (2020-01 cohort)	ACO 2 (2019-07 cohort)
Savings rate, PY2020, pp	3.4 (38)	5.6 (65)
Regional pull, pp	0.0 (26)	+2.3 (61)
Own-spending component, pp	−3.8 (41)	−4.7 (29)
Regional component, pp	+1.9 (44)	+3.0 (68)

The 2 early renewers already in ENHANCED in PY2021, for whom renewal could not raise the sharing rate, reported individually against the stayer distribution as the plan requires (section 5.6). Values with the stayers' percentile in parentheses. Exploratory; no test.

eTable 11. Per-capita spending by agreement year

Agreement start	Agreement year	Scheduled benchmark year	Group	ACOs, No.	Mean spending, $
2020	1	No	Other ACOs	135	10,609
2020	1	No	Early renewers	18	9,898
2020	2	No	Other ACOs	127	11,430
2020	2	No	Early renewers	18	10,880
2020	3	Yes	Other ACOs	118	11,860
2020	4	Yes	Other ACOs	96	12,450
2020	5	Yes	Other ACOs	73	13,206

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Agreement start	Agreement year	Scheduled benchmark year	Group	ACOs, No.	Mean spending, $
2022	1	No	Other ACOs	188	11,288
2022	1	No	Early renewers	18	10,931
2022	2	No	Other ACOs	164	11,912
2022	2	No	Early renewers	17	11,205
2022	3	Yes	Other ACOs	120	12,762
2022	3	Yes	Early renewers	15	11,892

Mean per-capita spending (US dollars) of assigned beneficiaries by agreement start year and agreement year (performance year - start year + 1), PY2020-PY2024 (plan section 5.6). Agreement years 3-5 are the scheduled benchmark years of the next agreement; early renewers out of a 2020 agreement are shown separately. Benchmark-year status cannot be separated from tenure here, so this is description only.

eMethods. Registration record and deviations

eTable 12. Registration record

Item	Value
Registry	OSF Registries, osf.io/c53w9, doi:10.17605/OSF.IO/C53W9
Registered	October 26, 2026, 13:22:16 UTC; public, no embargo
Plan	`PREREGISTRATION.md` version 0.6, SHA-256 `d0e374c8b493eb2f2fa7da1c0d8cbe46745a601233f11a81d2b2be73c402e8f2`
Code	commit `a4064f5`, archive SHA-256 `4c98d76a586151ab608851c69217a5ddb6b394201643a3de89a839bb1f659cd4`
Data	40 CMS public use files downloaded October 22, 2026, each with its SHA-256 hash in `data-snapshot.tsv`; archived on OSF[2]

Outcome lock. Every data file is hashed before it is read. A file whose hash matches a performance-year results file, a county spending file or the county assignment file could not be opened until the registration record existed, and the code had no bypass. Before registration, only the enrollment files, the CMS data dictionaries, the regulation text and the column names (not the values) of the outcome files were used.

Order of analysis. The stages ran in the order validate, power, confirm and robustness (concurrently; both depended only on validation and power), then H4, conventional and exploratory.

Each stage refused to start until its inputs existed. The robustness results were committed about 17 hours before the confirmatory results and were read only together with them, primary test first.

Deviations. All 7 stages ran with code identical to the registered commit. Two files changed after the last stage and produced no result: a reporting script for the online dashboard, which had shown PY2021 savings rates 100 times too large, and the lockfile `uv.lock`, which took security updates to 2 packages (filelock and virtualenv), neither of which enters an estimate. The lockfile is a registered file, so its change is logged as the one deviation; rerunning the analysis as registered uses the lockfile at the registered commit. One sentence in the plan's background (section 1.1) is imprecise: ENHANCED shares 75% of savings but up to 75% of losses, and the bar on moving from BASIC to ENHANCED within an agreement period is in the 2018 rule's preamble,[3] not in the regulation's text. Nothing in the analysis depends on it.

Reproducibility. A rerun from a fresh clone reproduced the validation stage exactly and every deterministic value of the power stage, but not the simulated values exactly: H1's simulated power was 77.6% against the committed 80.4%, and H3's detection shares 15% against 17% (pull) and 15% against 14% (BASIC level). The analysis reads its table without a fixed row order, so seeded random draws fall on ACOs in a different order in each run, and every resampled or simulated value varies within simulation error (about 1.6 points for a detection share from 500 datasets). No verdict is near a threshold at which that would matter.

STROBE checklist

Items of the STROBE statement for cohort studies,[4] with the place each is reported. "Supplement" refers to this document.

STROBE checklist

Item	No.	Recommendation (abbreviated)	Where reported
Title and abstract	1	Indicate the design; give an informative, balanced summary	Title (pre-registered analysis); Abstract (retrospective cohort study)
Background/rationale	2	Scientific background and rationale	Introduction, paragraphs 1-3
Objectives	3	Specific objectives and prespecified hypotheses	Introduction, paragraph 4; Methods, Hypotheses
Study design	4	Key elements of the design	Methods, Study design and preregistration

Item	No.	Recommendation (abbreviated)	Where reported
Setting	5	Setting, locations, dates of recruitment, exposure, follow-up, data collection	Methods, Data sources; Population and groups
Participants	6	Eligibility criteria, sources and methods of selection; follow-up	Methods, Population and groups; Figure 1
Variables	7	Define outcomes, exposures, predictors, confounders, effect modifiers	Methods, Variables; Hypotheses
Data sources/measurement	8	Sources and methods of measurement; comparability	Methods, Data sources; Variables (validation); eTable 1
Bias	9	Efforts to address potential sources of bias	Methods, Study design (preregistration, outcome lock), Statistical analysis (equivalence condition); Discussion, Limitations
Study size	10	How the study size was arrived at	Methods, Population and groups (all eligible ACOs); Power; eTable 2
Quantitative variables	11	How quantitative variables were handled; groupings	Methods, Variables; Statistical analysis (scaling by 2 SD)
Statistical methods	12a	All statistical methods, including for confounding	Methods, Statistical analysis
	12b	Methods for subgroups and interactions	None registered; single-cohort restrictions in eTables 3-4
	12c	How missing data were addressed	Methods, Variables (suppressed values); Results, Validation and sample
	12d	How loss to follow-up was addressed	Methods, Statistical analysis (not yet renewed; balanced panel); eTable 8
	12e	Sensitivity analyses	Methods, Covariate-adjusted estimates, conventional methods and robustness; eTables 3-6
Participants	13a	Numbers at each stage	Figure 1; Results, Validation and sample
	13b	Reasons for nonparticipation at each stage	Figure 1; Results, Validation and sample
	13c	Flow diagram	Figure 1
Descriptive data	14a	Characteristics of participants; exposures and confounders	Table 1 (group, cohort, track, case mix, savings rate, constructed variables); the H3 adjustment factors (size, revenue category, assignment method, agreement number) are not in the results files and are not tabulated

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Item	No.	Recommendation (abbreviated)	Where reported
	14b	Number with missing data for each variable	Table 1; Figure 1
	14c	Follow-up time	eTable 8 (mean included post-period years by group)
Outcome data	15	Numbers of outcome events or summary measures over time	Early renewal: 29 of 306 compared ACOs (Table 1), 24 of 206 in H3 (Table 3); spending over time: eTable 11
Main results	16a	Unadjusted and adjusted estimates with precision	Tables 2-4; Figures 2-4; eTables 3, 6
	16b	Category boundaries for continuous variables	Not applicable (continuous variables not categorized)
	16c	Translation of relative risk into absolute risk	Results (odds multiplier; dollar translation); eTable 5 (observed and predicted shares by decile)
Other analyses	17	Subgroups, interactions and sensitivity analyses	Results, Robustness and conventional methods; eTables 3-11
Key results	18	Summarize key results with reference to objectives	Discussion, paragraph 1
Limitations	19	Sources of potential bias or imprecision; direction and magnitude	Discussion, Limitations
Interpretation	20	Cautious overall interpretation	Discussion; Conclusions
Generalisability	21	External validity	Discussion, Implications (2024 rules); Limitations
Funding	22	Funding and role of funders	Declarations

References

1. Harrold ST. Why did Medicare ACOs reset their benchmarks early? A pre-registered analysis of early renewal in the Medicare Shared Savings Program. Published online 2026. doi:10.17605/OSF.IO/C53W9

2. Harrold ST. Why did Medicare ACOs reset their benchmarks early? Project files: Results notebook exports and archived CMS data snapshot. Published online 2026. doi:10.17605/OSF.IO/Z4AN5

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3. Centers for Medicare & Medicaid Services. Medicare program; Medicare Shared Savings Program; accountable care organizations: Pathways to Success and extreme and uncontrollable circumstances policies for performance year 2017. Final rule. Published online 2018. https://www.federalregister.gov/documents/2018/12/31/2018-27981/medicare-program-medicare-shared-savings-program-accountable-care-organizations-pathways-to-success

4. von Elm E, Altman DG, Egger M, et al. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies. *PLoS Medicine*. 2007;4(10):e296. doi:10.1371/journal.pmed.0040296

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